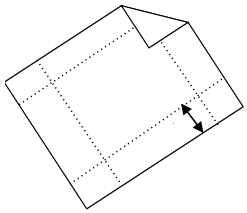
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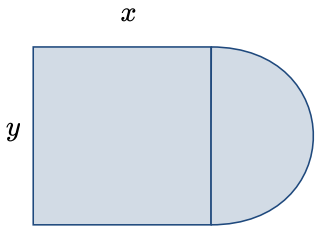
Optimisation Problems/Modelling

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We have a sheet of A4 paper, which we want to fold into a cuboid. What height should we choose for the cuboid in order to maximise the volume?

These are examples of **optimisation problems**: we're trying to maximise/minimise some quantity by choosing an appropriate value of a variable that we can control.



We have 50m of fencing, and want to make a bear pen of the following shape, such that the area is maximised. What should we choose x and y to be?

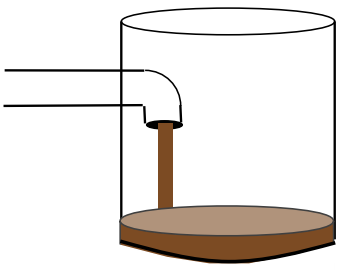
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'Rate of change'

71

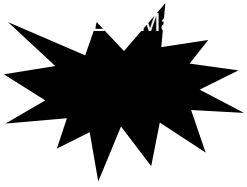
Up to now we've had y in terms of x, where $\frac{dy}{dx}$ means "the **rate** at which y changes with respect to x".

But we can use similar gradient function notation for other **physical quantities**.



A sewage container fills at a rate of 20 cm^3 per second.

How could we use appropriate notation to represent this?



1.38 Applications of Calculus

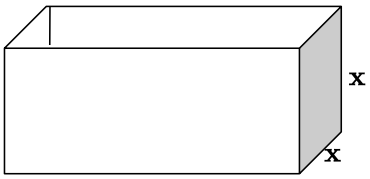
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Example Optimisation Problem

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Optimisation problems in an exam usually follow the following pattern:

- There are 2 variables involved (you may have to introduce one yourself), typically lengths.
- There are expressions for **two different physical quantities**:
  - One is a **constraint**, e.g. "the surface area is 20cm^2".
  - The **other we wish to maximise/minimise**, e.g. "we wish to maximise the volume".
- We use the constraint to **eliminate one of the variables** in the latter equation, so that it is then **just in terms of one variable**, and we can then use differentiation to find the turning point.



[Textbook] A large tank in the shape of a cuboid is to be made from 54m^2 of sheet metal. The tank has a horizontal base and no top. The height of the tank is x metres. Two of the opposite vertical faces are squares.

a) Show that the volume, V m^3, of the tank is given by $V = 18x - \frac{2}{3}x^3$.

b) Given that x can vary, use differentiation to find the maximum or minimum value of V.

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Test Your Understanding

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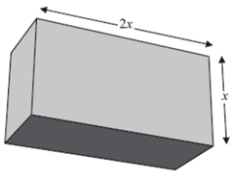


Figure 2

A cuboid has a rectangular cross-section where the length of the rectangle is equal to twice its width, x cm, as shown in Figure 2.

The volume of the cuboid is 81 cubic centimetres.

(*a*) Show that the total length, L cm, of the twelve edges of the cuboid is given by

$$L = 12x + \frac{162}{x^2}.$$

(3)

(*b*) Use calculus to find the minimum value of L.

(6)

(*c*) Justify, by further differentiation, that the value of L that you have found is a minimum.

(2)

(a)

(b)

(c)

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6 A rectangular garden is fenced on three sides, and the house forms the fourth side of the rectangle.

a Given that the total length of the fence is $80\,\text{m}$, show that the area, A, of the garden is given by the formula $A = y(80 - 2y)$, where y is the distance from the house to the end of the garden.

b Given that the area is a maximum for this length of fence, find the dimensions of the enclosed garden, and the area which is enclosed.

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7 A closed cylinder has total surface area equal to 600π.
a Show that the volume, $V\,\text{cm}^3$, of this cylinder is given by the formula $V = 300\pi r - \pi r^3$, where $r\,\text{cm}$ is the radius of the cylinder.
b Find the maximum volume of such a cylinder.

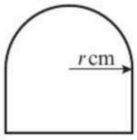
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9 A shape consists of a rectangular base with a semicircular top, as shown.



a Given that the perimeter of the shape is 40 cm, show that its area, A cm^2, is given by the formula

$$A = 40r - 2r^2 - \frac{\pi r^2}{2}$$

where r cm is the radius of the semicircle. **(2 marks)**

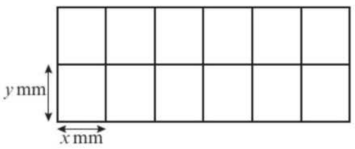
b Hence find the maximum value for the area of the shape. **(4 marks)**

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10 The shape shown is a wire frame in the form of a large rectangle split by parallel lengths of wire into 12 smaller equal-sized rectangles.



a Given that the total length of wire used to complete the whole frame is $1512\,\text{mm}$, show that the area of the whole shape, $A\,\text{mm}^2$, is given by the formula

$A = 1296x - \frac{108x^2}{7}$

where x mm is the width of one of the smaller rectangles. \hfill (4 marks)

b Hence find the maximum area which can be enclosed in this way. \hfill (4 marks)

1.38 Applications of Calculus

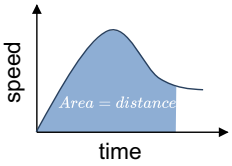
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Definite Integration

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So far we've seen integration as '**the opposite of differentiation**', allowing us to find $y = f(x)$ when we know the gradient function $y = f'(x)$.

In practical settings however the most useful use of integration is that **it finds the area under a graph**. Remember at GCSE for example when you estimated the area under a speed-time graph, using trapeziums, to get the distance?
If you knew the equation of the curve, you could get the exact area!

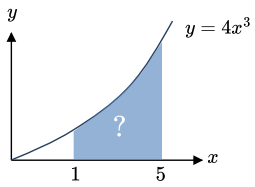


Before we do this, we need to understand how to find a **definite integral**:

These are known as **limits**, which give the values of x we're finding the area between.

We integrate as normal, but put expression in **square brackets**, meaning **we still need to evaluate the integrated expression using the limits**.

$$\int_1^5 4x^3\,dx =$$



Write $(\dots) - (\dots)$ and evaluate the expression for each of the limits, top one first.

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Another Example

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You can use the $\left[\!\left[\int_b^a \square\right]\!\right]$ button on your calculator to evaluate definite integrals.
But only use it to check your answer.

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Problem Solving

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Given that P is a constant and $\int_1^5 (2Px + 7)dx = 4P^2$, show that there are two possible values for P and find these values.

Remember: P is a constant, so just treat it as a number.

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4 Given that A is a constant and $\int_1^4 (6\sqrt{x} - A)\,dx = A^2$, show that there are two possible values for A and find these values.

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5 Use calculus to find the value of $\int_1^9 (2x - 3\sqrt{x})\,\mathrm{d}x$.

6 Evaluate $\int_4^{12} \frac{2}{\sqrt{x}}\,\mathrm{d}x$, giving your answer in the form $a + b\sqrt{3}$, where a and b are integers.

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7 Given that $\int_1^k \frac{1}{\sqrt{x}}\,\mathrm{d}x = 3$, calculate the value of k.